Fashion-MNIST Generative Modelling — Analysis & Short Report

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Comparisons

AE vs VAE (deterministic vs probabilistic)

The Task 2 Denoising Autoencoder (DAE) learns a single deterministic code per image and is optimised purely for reconstruction (MSE); its latent space has no enforced structure, so sampling or interpolating in it is unreliable. The Task 3 VAE instead learns a distribution q(z|x) = N(mu, sigma^2) per image and is regularised (via the KL term) toward a standard normal prior, at the cost of blurrier reconstructions. That regularisation is exactly what buys the VAE the ability to sample from scratch and interpolate smoothly between real images — capabilities the DAE's latent space does not reliably support.

VAE vs cGAN (likelihood vs adversarial)

The VAE is trained to maximise a tractable lower bound on data likelihood (ELBO); its pixel-wise reconstruction term rewards "safe" averaged pixel values under uncertainty, producing systematically blurrier samples. The cGAN never computes a likelihood at all — the generator is trained purely to fool a discriminator, so there is no pixel-averaging pressure, and its outputs are visibly sharper. The tradeoff is optimisation stability: the VAE's loss is a single well-behaved objective that improves monotonically, while the cGAN's adversarial min-max game needed label smoothing and careful learning-rate/beta tuning to avoid the discriminator overpowering the generator.

Why labels improve control in cGAN

An unconditional GAN can sample a garment but gives no lever to request which garment. Concatenating a label embedding into the generator's input, and a label map into the discriminator's input, turns the single unconditional data distribution into 10 class-conditional sub-distributions the discriminator can hold the generator accountable to per class — "this doesn't look like a real sandal" is a strictly stronger training signal than "this doesn't look like a real image." That is what makes explicit per-class sampling possible at all, which is not available to the plain VAE or DAE without retrofitting a similar conditioning mechanism.

Takeaways

What each model captures well or poorly

The DAE is the strongest pure compressor/denoiser but the weakest generator, since its latent space is not meant for sampling. The VAE is the best-behaved generative model to train and gives a genuinely useful, interpolable latent space, at the cost of blur. The cGAN gives the sharpest, most controllable samples but is the most finicky to train and offers no direct latent-interpolation guarantee the way the VAE does. AdaIN-based style mixing (Task 5) is not a generator on its own — it is a mechanism for recombining representations from an existing encoder/decoder, and its quality is capped by how well that encoder/decoder was trained in the first place.

Practical notes (learning rate, noise, batch size, etc.)

Gaussian noise sigma≈0.3 was strong enough to force the DAE to genuinely denoise rather than pass an easy identity mapping through. The VAE's higher learning rate (2e-3 vs the DAE's 1e-3) trained stably only because the small bottleneck (20-dim) and BCE+KL objective are both well-conditioned; a larger latent dimension would likely need a lower rate or KL warm-up to avoid posterior collapse. For the cGAN, real-label smoothing (0.9 instead of 1.0) and the lower, momentum-adjusted Adam betas (0.5, 0.999) were both necessary in practice — without them the discriminator tends to converge too quickly early in training and starve the generator's gradient signal. Batch size 128 was a good stability/throughput tradeoff for all three trained models on this dataset size. All training for this submission ran on CPU; the identical code is GPU-ready via automatic device detection and would train substantially faster on a Colab GPU runtime.